AKSHAT CHAWLA Backend AI/ML Engineer | Node.js | Python | System Design

+91 9897976962 | akshatchawla1997@gmail.com | github.com/akshatchawla1997 | linkedin.com/in/akshat-sd1/

Backend & AI/ML Engineer with 6+ years of experience designing scalable systems and intelligent applications. Strong in Python, Node.js, PostgreSQL, and cloud infrastructure, with hands-on work in system design, performance optimization, and AI-powered backend services. Focused on building reliable, production-grade architectures at scale.

SKILLS

Languages	Python, Node.js, SQL
Backend	FastAPI, Express.js, REST APIs, WebSockets, Microservices, PostgreSQL, Redis
AI / ML	PyTorch, TensorFlow, Scikit-learn, NLP, Computer Vision, Model Inference, Data Processing, Agentic AI, Orchestration, RAG(Retrieval-Augmented Generation, LangChain/LlamaIndex, Vector Databases (Pinecone/Milvus)
DevOps / Cloud	Docker, AWS (EC2, S3, Lambda, IAM), GCP (GKE), CI/CD, Linux, Git, Kubernetes (K8s), Jenkins

WORK EXPERIENCE

AI & Backend Developer, *Aonetech* Dec 2024 – Present

- Architected a high-concurrency microservices backend using Node.js and Python, implementing Redis-based caching and PostgreSQL optimization to handle complex data workflows.
- Integrated AI/ML inference pipelines into production APIs, including image classification models and LLM-powered workflows using Gemini and OpenAI for real-time decision systems.
- Engineered a custom AI Orchestration layer that utilizes NLU to map unstructured user intent to automated backend actions, facilitating dynamic itinerary generation.

Strategic Backend & AI Developer (Contractual), *(Momentus | Mobrilz | Plutos One)* | Onsite (Gurgaon, India) Dec 2023 - Dec 2024

- AI/ML Execution: Developed intelligent Rasa NLU chatbots integrated with Google My Business for automated lead generation.
- DevOps & Scaling: Containerized applications with Docker & established CI/CD pipelines via GitHub Actions & Jenkins.
- Performance: Optimized PostgreSQL indexing and Redis caching strategies, reducing API response times across high-traffic e-commerce services.

PISC, *Backend Developer* | Noida, India July 2022 - Dec 2023

- Built and maintained scalable backend services using Node.js, Python (Django), ensuring high performance & reliability.
- Integrated a wide range of third-party services including payment gateways, authentication providers into PWA.
- Implemented real-time communication features using WebSocket & Socket.IO to enhance UX in live data applications.

Resonance International Inc., *Operations & Technical Associate* Dec 2019 – March 2022

- Optimized internal data reporting workflows & maintained 100% operational continuity during the 2020-2021 pandemic period.

PROJECTS

Map My Tour Conversational AI Travel Assistant **| PostgreSQL, GCP, PyTorch, TensorFlow** Jan 2026 - present

- Designed NLP pipeline to understand user queries and map them to structured travel plans.
- Developed backend orchestration services to fetch, process, and serve travel data through REST APIs.
- Created a wrapper orchestration layer over & Gemini APIs to route queries and optimize responses for travel planning.

MultiFolks | E-commerce Platform with AI Image Processing **| Python, FastAPI, PostgreSQL, Gemini API, Docker** Sept 2025 - Feb 2026

- Developed the backend for an e-commerce platform handling product management, orders, and user workflows using Node.js and PostgreSQL.
- Built an AI image processing and classification pipeline using Gemini API and deep learning models, integrated with FastAPI services for real-time inference after preprocessing images to remove facial obstructions (e.g., glasses).
- Containerized the ML and API services using Docker for consistent deployment and scalability.

Ticketing NFT Decentralized Web3 Event Ticketing Platform **| Node.js, Express.js, PostgreSQL, JavaScript, AWS, Web3.js, IPFS, Smart Contracts** April 2025 – June 2025

- Developed an NFT-based event ticketing platform that mints, sells, and verifies tickets on the blockchain to ensure authenticity, security, and prevent fraud/scalping.
- Integrated Web3 wallet authentication and smart contract interactions for ticket minting, transfer restrictions, and royalty configurations.
- Enabled real-time ticket validation via QR scanning and blockchain ownership verification, ensuring smooth on-ground event entry operations.

Education

MCA, Graphic Era University, Uttarakhand 2017 – 2019